

# LESSON 6.3

## DETERMINING PROPORTIONAL RELATIONSHIPS



### LESSON INTRODUCTION

**MA.8.AR.3.1** Determine if a linear relationship is also a proportional relationship.

### LEARNING TARGETS

- I can determine if a linear relationship is also a proportional relationship given a graph, table, equation, or written description.

### BENCHMARK COHERENCE

#### BUILDING ON

**MA.7.AR.4.1** Determine whether two quantities have a proportional relationship by examining a table, graph, or written description.

**MA.7.AR.4.4** Given any representation of a proportional relationship, translate the representation to a written description, table, or equation.

#### WORKING TOWARD

N/A

### ESSENTIAL QUESTIONS

- How do you determine whether a graph, table, or equation is proportional?

### LESSON NARRATIVE

Using their knowledge of proportional relationships, students explore the difference between proportional and non-proportional relationships in equations, graphs, and tables. Students create each of these from written descriptions and then explore and analyze their results.

### COMPONENT SUMMARY

#### 6.3.1 WARM-UP (~5 MIN.)

Students decide which scenario they would prefer

#### 6.3.2 EXPLORATION (20 MIN.)

Explore proportional versus non-proportional relationships

#### 6.3.3 GUIDED PRACTICE (15 MIN.)

Practice determining if relationships are proportional

#### 6.3.4 WRAP-UP (~5 MIN.)

Determine if equations, tables, and graphs represent proportional relationships

## RESOURCES

### GLOSSARY

Key Terms: N/A

Additional Terms:

- Proportional Relationship
- Independent variable
- Dependent variable
- Constant of proportionality
- Equivalent
- Ratios

### MATERIALS

- (Optional) Chart paper & markers
- (Optional) Calculators

### ADDITIONAL PREPARATION

- Consider having the graph of  $y = \frac{3}{x}$  prepared to display for the class.

## TEACHER TIPS

### COMMON MISCONCEPTIONS

- Students may not realize that start-up or service fees are one-time payments.
- Students may confuse the initial value and the rate of change.

### THINGS TO CONSIDER

- Consider creating and displaying a Word Wall with vocabulary definitions, examples, and visuals.

### LITERACY AND LANGUAGE ROUTINES

#### Think-Pair-Share

In 6.3.2 Exploration, have students initially complete the problems independently. Then, ask students to discuss their work with a partner. Finally, debrief with the whole class. These discussions will help students gain confidence and understanding.

### MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

#### MA.K12.MTR.2.1 Multiple Representations

In 6.3.3 Guided Practice, students represent proportional relationships as a table, graph, and an equation. Having students work with multiple representations of proportional relationships deepens their conceptual understanding and enhances their procedural fluency.

## DIFFERENTIATE INSTRUCTION

In 6.3.3 Guided Practice, consider reminding students the equation of a proportional relationship is  $y=kx$ . For students struggling to tell if a graph represents a proportional relationship, students can substitute 0 for  $x$ , if the output is not also 0, then the graph is not proportional.

## 6.3.1 WARM-UP: WOULD YOU RATHER?

### TEACHER MOVES

Ask students if they agree with students who share their selection and why. Ask students to share their experiences selling items (i.e., a lemonade stand) and how they determined the price. Consider inviting a business owner to join class to discuss how they determine their prices.



### 6.3.1 Warm-Up: Would You Rather?

Consider the following situations:

#### Situation A:

The cost to make 36 cupcakes is \$10. You charge \$0.50 per cupcake.

#### Situation B:

The cost to make 36 cupcakes is \$8.00. You charge \$18.00 for all three dozen cupcakes.



- Which situation would you rather be in?  
*Situation B. Possible Responses: I would rather be in Situation B because I would make a profit of \$10.00. In Situation A the profit would only be \$8.00. I would make \$2.00 more from Situation B.*
- Explain or show your work to justify your reasoning.  
 A.  $36 \times .50 = \$18.00$  then  $\$18.00 - \$10.00 = \$8.00$   
 B.  $\$18.00 - \$8.00 = \$10.00$   
*Possible Responses: I would rather be in Situation B because I would make a profit of \$10.00. In Situation A the profit would only be \$8.00. I would earn \$2.00 more from Situation B.*



### 6.3.2 Exploration: Exploring Proportional vs. Non-Proportional Relationships

Work with your partner to complete the following.

1. Diego earns \$10 per hour babysitting. He has no money saved before he starts babysitting and plans to save all of his earnings. Complete the table with Diego's savings.

| Hours Worked | Amount Saved |
|--------------|--------------|
| 0            | 0            |
| 1            | 10           |
| 2            | 20           |
| 4            | 40           |
| 6.5          | 65           |
| 11           | 110          |
| $x$          | $10x$        |

## 6.3.2 EXPLORATION: EXPLORING PROPORTIONAL VS. NON-PROPORTIONAL RELATIONSHIPS

### TEACHER MOVES

#### Think-Pair-Share

Have students initially complete the problems independently. Then, ask students to discuss their work with a partner. Finally, debrief with the whole class. These discussions will help students gain confidence and understanding.

2. Diego's sister, Camilla, started with \$30 saved before she starts babysitting. She earns \$10 per hour as well and saves all that she earns. Complete the table with Camilla's savings.

| Hours Worked | Amount Saved |
|--------------|--------------|
| 0            | 30           |
| 1            | 40           |
| 2            | 50           |
| 4            | 70           |
| 6.5          | 95           |
| 11           | 140          |
| $x$          | $10x + 30$   |

### DIFFERENTIATE INSTRUCTION

Consider having students add an additional column titled "constant of proportionality" where they calculate the ratio of  $y$  to  $x$ . This column will help students see that the ratios are not equivalent.

## 6.3.2 EXPLORATION: EXPLORING PROPORTIONAL VS. NON-PROPORTIONAL RELATIONSHIPS (CONTINUED)

### TEACHER MOVES

#### Think-Pair-Share

Have students initially complete the problems independently. Then, ask students to discuss their work with a partner. Finally, debrief with the whole class. These discussions will help students gain confidence and understanding.

### ENRICHMENT

#### Notice and Wonder

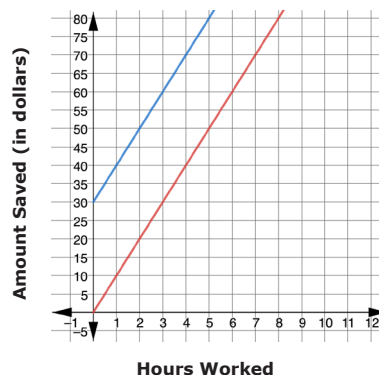
What do you notice about the equation of Diego's savings? What do you notice about the equation of Camilla's savings? What do you wonder?

### TEACHER MOVES

Prompt students to look at the original descriptions of Diego's and Camilla's savings. Ask students what they can look for in the description to determine if the relationship is proportional.

Diego's savings is modeled by the **red line**.

Camilla's savings is modeled by the **blue line**.



3. Using evidence from the tables and the graph, explain which line represents a proportional relationship.  
*Answers will vary. Diego's savings is a proportional relationship because its graph is a straight line that passes through the origin. Camilla's savings is not proportional because it starts at (0,30) and does not pass through the origin.*

4. Graphs of proportional relationships meet two criteria. Complete the statements to describe both criteria.
- The graph must be **straight**.
  - The graph must cross the coordinates **(0,0)**.

The equations used to model the savings for Diego and Camilla are shown.

| Diego's Savings | Camilla's Savings |
|-----------------|-------------------|
| $y = 10x$       | $y = 10x + 30$    |

5. Discuss with your partner the difference between the two equations and explain how to determine if an equation represents a proportional relationship or not.  
*Answers will vary. Diego's equation multiplies  $x$  by 10 each time so if  $x = 0$ , then  $y = 10 \cdot 0$ ;  $y$  would also equal 0. That would make it a proportional relationship. Camilla's equation multiplies  $x$  by 10 each time but then adds 30. In her equation, when  $x$  is equal to zero  $y$  is equal to 30 so the graph does not go through the origin. It is not proportional.*



### 6.3.3 Guided Practice: Determining if a Relationship is Proportional

- Identify if each equation is proportional or not.

If it is proportional, circle the constant of proportionality in the equation. If it is not proportional, box the portion of the equation that indicates the relationship is not proportional.

| Equation                       | Proportional or Not Proportional?                                                       |
|--------------------------------|-----------------------------------------------------------------------------------------|
| $y = \boxed{8} + x$            | <input type="radio"/> proportional<br><input checked="" type="radio"/> not proportional |
| $y = \boxed{-4}x$              | <input checked="" type="radio"/> proportional<br><input type="radio"/> not proportional |
| $y = \boxed{5} + \frac{1}{2}x$ | <input type="radio"/> proportional<br><input checked="" type="radio"/> not proportional |
| $y = \boxed{\frac{1}{5}}x$     | <input checked="" type="radio"/> proportional<br><input type="radio"/> not proportional |
| $y = \boxed{6} - 2x$           | <input type="radio"/> proportional<br><input checked="" type="radio"/> not proportional |
| $y = \boxed{-0.75}x$           | <input checked="" type="radio"/> proportional<br><input type="radio"/> not proportional |
| $y = \frac{3}{\boxed{x}}$      | <input type="radio"/> proportional<br><input checked="" type="radio"/> not proportional |

- Jada's grandparents started a savings account for her in 2010. The table shows the amount in the account each year.

| Years Since 2010 | Amount (in dollars) |
|------------------|---------------------|
| 0                | 600                 |
| 2                | 750                 |
| 4                | 900                 |
| 6                | 1050                |

Explain whether this is a proportional relationship or not.

*Not Proportional. Possible Responses: When the  $x$  value equals zero, the  $y$  value equals 600. This means that the line that is graphed will not pass through  $(0,0)$ , so it is not a proportional relationship. The equation is  $y = 75x + 600$ . The equation needs to be in the form of  $y = kx$  to be proportional.*

## 6.3.3 GUIDED PRACTICE: DETERMINING IF A RELATIONSHIP IS PROPORTIONAL

### DIFFERENTIATE INSTRUCTION

Consider a reminder of the equation of a proportional relationship  $y=kx$ . For students struggling to tell if the equations are proportional, explain that they can substitute 0 for  $x$ . If the output,  $y$ , is not also 0, then the graph is not proportional.

### TEACHER MOVES

The equation  $y=3/x$  will be difficult for some students. Explain that if they can substitute 0 for  $x$ , the output,  $3/0$ , is undefined. It may be helpful for students to see the graph of  $y=3/x$ . Consider having the graph prepared to display for the class.

### ENRICHMENT

- What does the point  $(0,600)$  represent in the context of this problem?
- How does it help us determine if the graph is proportional or not?

### 6.3.3 GUIDED PRACTICE: DETERMINING IF A RELATIONSHIP IS PROPORTIONAL (CONTINUED)

#### ENRICHMENT

How will the membership fee impact the total amount members spend? What is a membership fee?

#### DIFFERENTIATE INSTRUCTION

Consider a review of start-up fees and what they mean in the context of this situation. If there is a start-up fee, will the relationship be proportional? Why or why not?

#### QUESTIONING

To extend students thinking, prompt them to write an equation to represent the relationship between the number of days Ashley has read and the number of pages read.

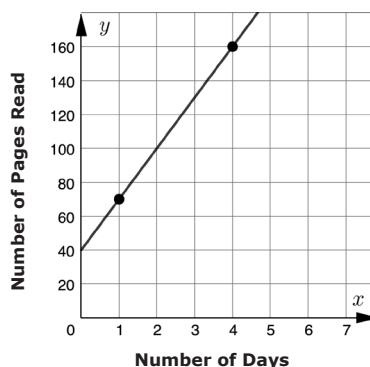
3. Customers at the local gym pay a \$59.99 membership fee to join and then a \$15.50 fee for each class they attend.
- a. Complete the table with the total costs per classes attended.

| Classes Attended | Total Cost |
|------------------|------------|
| 0                | 59.99      |
| 1                | 75.49      |
| 2                | 90.99      |
| 3                | 106.49     |
| 4                | 121.99     |
| 5                | 137.49     |
| 6                | 152.99     |
| 7                | 168.49     |

- b. Use evidence from the written description or the table to explain if the situation is a proportional relationship or not.

*Non-proportional. Possible Responses: The customers must pay \$59.99 before they actually attend the classes. This is added to the monthly fee, so the relationship is not proportional. The table shows that when  $x = 0$  then  $y = 59.00$ . The graph of the line will not pass through the origin, and therefore, the relationship is non-proportional.*

4. Ashley is reading a book for book club. After reading the first 40 pages of the book in a week, she plans to read 30 pages each day until she finishes to complete the book in time for her next book club meeting. The graph of the situation is shown.



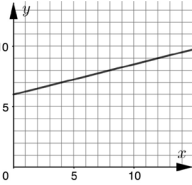
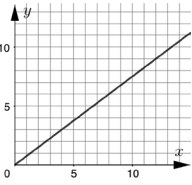
Ashley's friend said the situation is proportional because she reads the same number of pages each day and the graph is a straight line. Use the graph to help correct her friend.

*Answers will vary. The graph starts at (0,40) and does not go through the origin. The graph of a proportional relationship is a straight line that passes through the origin.*



6.3.4 Wrap-Up: Ticket Out the Door

1. Mark each relationship as either proportional or not proportional.

| Relationship                                                                                                                        | Proportional or Not Proportional?                                            |   |   |   |   |   |   |   |                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---|---|---|---|---|---|---|------------------------------------------------------------------------------|
| $y = 1.7x$                                                                                                                          | <input type="radio"/> proportional<br><input type="radio"/> not proportional |   |   |   |   |   |   |   |                                                                              |
|                                                    | <input type="radio"/> proportional<br><input type="radio"/> not proportional |   |   |   |   |   |   |   |                                                                              |
| $y = \frac{3}{2}x$                                                                                                                  | <input type="radio"/> proportional<br><input type="radio"/> not proportional |   |   |   |   |   |   |   |                                                                              |
| <table><tr><th>x</th><th>y</th></tr><tr><td>3</td><td>1</td></tr><tr><td>6</td><td>2</td></tr><tr><td>9</td><td>3</td></tr></table> | x                                                                            | y | 3 | 1 | 6 | 2 | 9 | 3 | <input type="radio"/> proportional<br><input type="radio"/> not proportional |
| x                                                                                                                                   | y                                                                            |   |   |   |   |   |   |   |                                                                              |
| 3                                                                                                                                   | 1                                                                            |   |   |   |   |   |   |   |                                                                              |
| 6                                                                                                                                   | 2                                                                            |   |   |   |   |   |   |   |                                                                              |
| 9                                                                                                                                   | 3                                                                            |   |   |   |   |   |   |   |                                                                              |
| $y = 2x - 7$                                                                                                                        | <input type="radio"/> proportional<br><input type="radio"/> not proportional |   |   |   |   |   |   |   |                                                                              |
|                                                    | <input type="radio"/> proportional<br><input type="radio"/> not proportional |   |   |   |   |   |   |   |                                                                              |

6.3.4 WRAP-UP: TICKET OUT THE DOOR

QUESTIONING

Ask students to write a justification of why each representation is proportional or not proportional.